



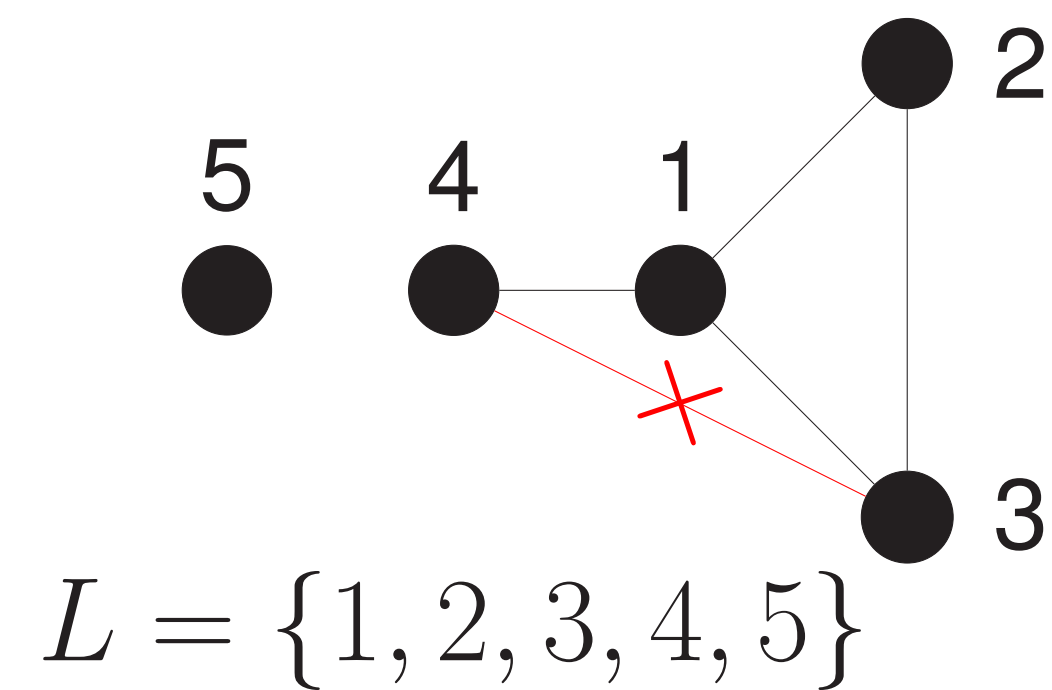
On The Spum and Sum-Diameter of Paths

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Sum Graphs (Harary, '90)

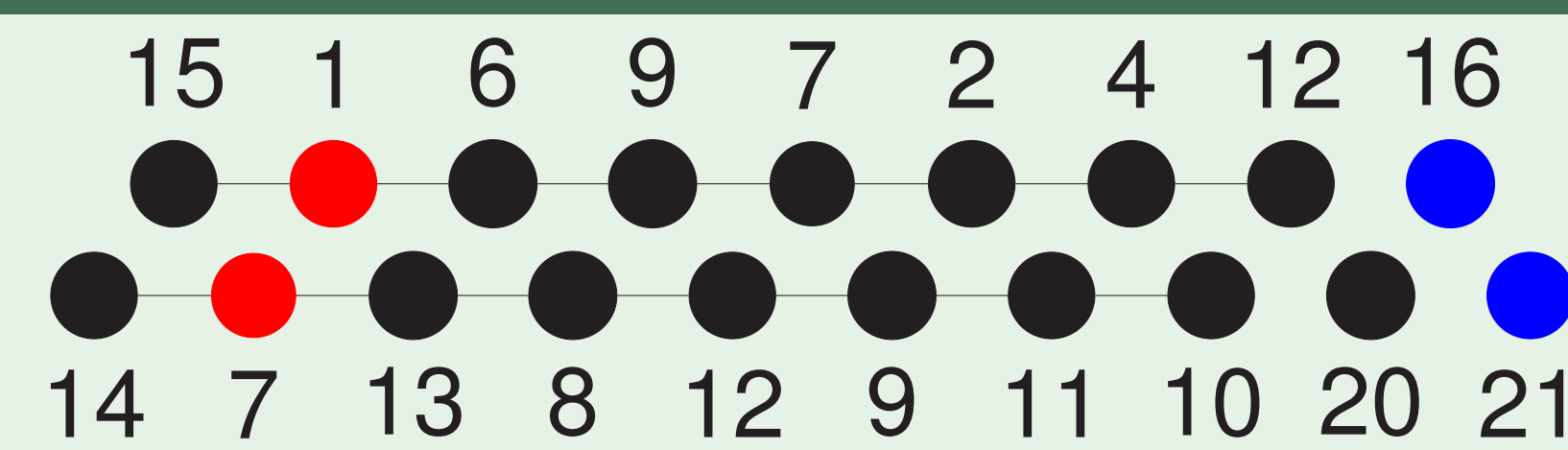
- The *sum graph* $G(V, E)$ with *sum graph labeling* $L \subseteq \mathbb{Z}^+$ satisfies $V = L$ and $(u, v) \in E$ if and only if $u + v \in L$.



Sum-Diameter (Li '22)

- Let $\text{sd}(G)$ be the minimum $\text{range}(L)$ over all sum graph labelings L of the union of graph G and any number of isolated vertices.

Example: $\text{sd}(P_8) = 14 < \text{spum}(P_8) = 15$



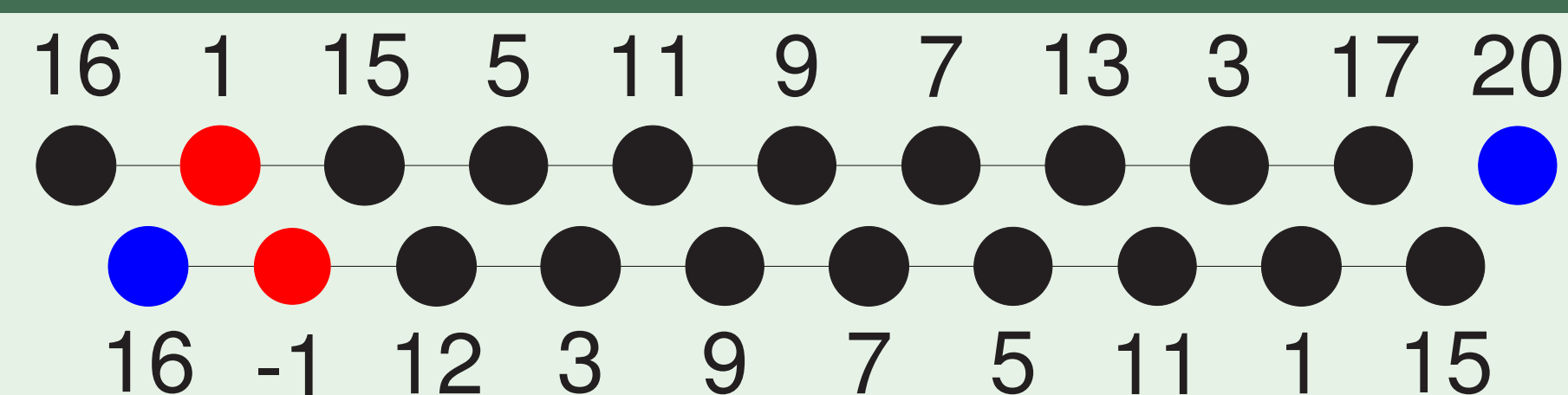
Integral Spum $\text{ispum}(G)$ (Singla, Tiwari, Tripathi '21)

- Let an *integral sum graph* be a sum graph with integer labels.
- Let $\zeta(G)$ be the minimum number such that the union of G and $\zeta(G)$ isolated vertices is an integral sum graph.
- Let $\text{ispum}(G)$ be the minimum $\text{range}(L)$ over all integral sum graph labelings L of the union of G and $\zeta(G)$ isolated vertices.

Natural Questions for Integral Spum

- Is it possible that $\zeta(G) = 0$ for a connected graph G ? **Yes!**
- Is it possible that $\text{ispum}(P_n) < \text{spum}(P_n)$? **Yes!**

Example: $\text{ispum}(P_{10}) = 17 < \text{spum}(P_{10}) = 19$



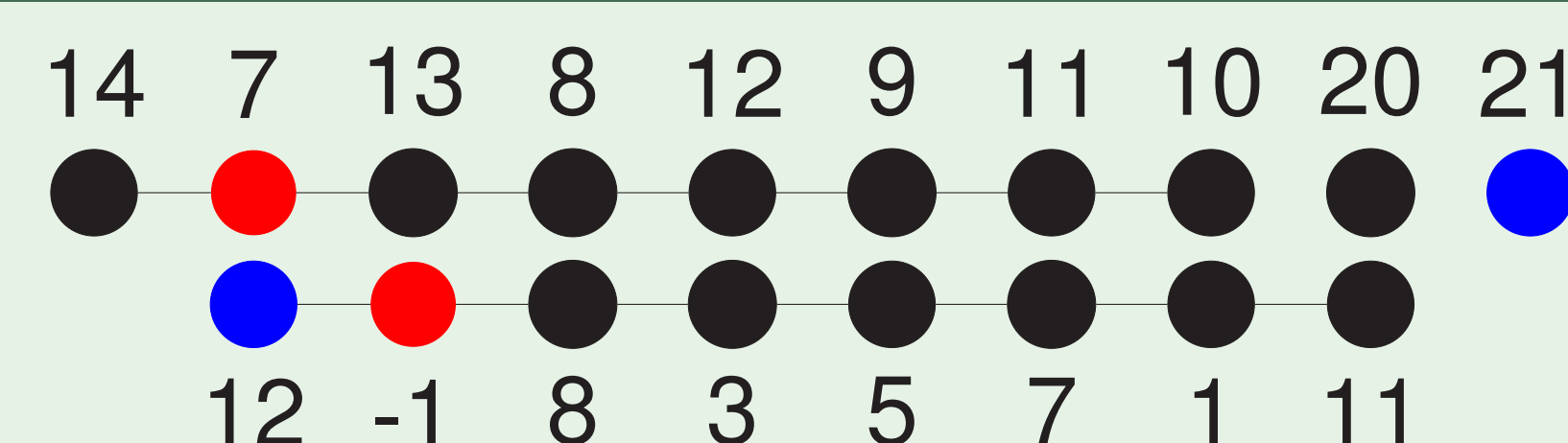
Integral Sum-Diameter $\text{isd}(G)$ (Li '22)

- Let $\text{isd}(G)$ be the minimum $\text{range}(L)$ over all integral sum graph labelings L of the union of G and any number of isolated vertices.

Natural Questions for Integral Sum-Diameter

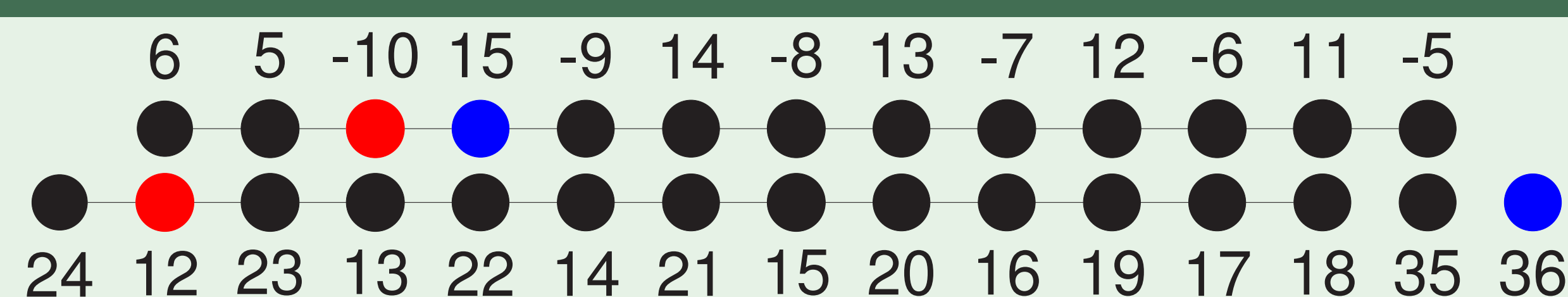
- Is it possible that $\text{isd}(P_n) < \text{sd}(P_n)$? **Yes!**

Example: $\text{isd}(P_8) = 13 < \text{sd}(P_8) = 14$



- Is it possible that $\text{isd}(P_n) < \text{ispum}(P_n)$? **Yes!**

Example: $\text{isd}(P_{13}) = 24 < \text{ispum}(P_{13}) = 25$



Theorem 2: Sum-Diameter

Theorem (Li, '22)

- If $n \geq 3$, then $2n - 3 \leq \text{sd}(P_n) \leq 2n - 2$.

Theorem (B.C.T., '23)

- If $n \geq 7$, then $\text{sd}(P_n) = 2n - 2$.

Theorem 3: Integral Spum

Theorem (Singla, Tiwari, Tripathi '21)

- If $n \geq 7$, then

$$2n - 5 \leq \text{ispum}(P_n) \leq \begin{cases} 2n - 3 & \text{if } n \text{ is even} \\ \frac{5}{2}(n - 3) & \text{if } n \text{ is odd} \end{cases}$$

Theorem (B.C.T., '23)

- If $n \geq 7$, then

$$2n - 3 \leq \text{ispum}(P_n) \leq \begin{cases} 2n - 3 & \text{if } n \text{ is even} \\ \frac{5}{2}(n - 3) & \text{if } n \text{ is odd} \end{cases}$$

Theorem 4: Integral Sum-Diameter

Theorem (Li, '22)

- If $n \geq 3$, then

$$2n - 5 \leq \text{isd}(P_n) \leq \begin{cases} 2n - 2 & \text{if } n \text{ is odd} \\ 2n - 3 & \text{if } n \text{ is even} \end{cases}$$

Theorem (B.C.T., '23)

- If $n \geq 10$, then

$$\text{isd}(P_n) = \begin{cases} 2n - 3 & \text{if } n \text{ is even} \\ 2n - 2 & \text{if } n \text{ is odd} \end{cases}$$

Acknowledgments

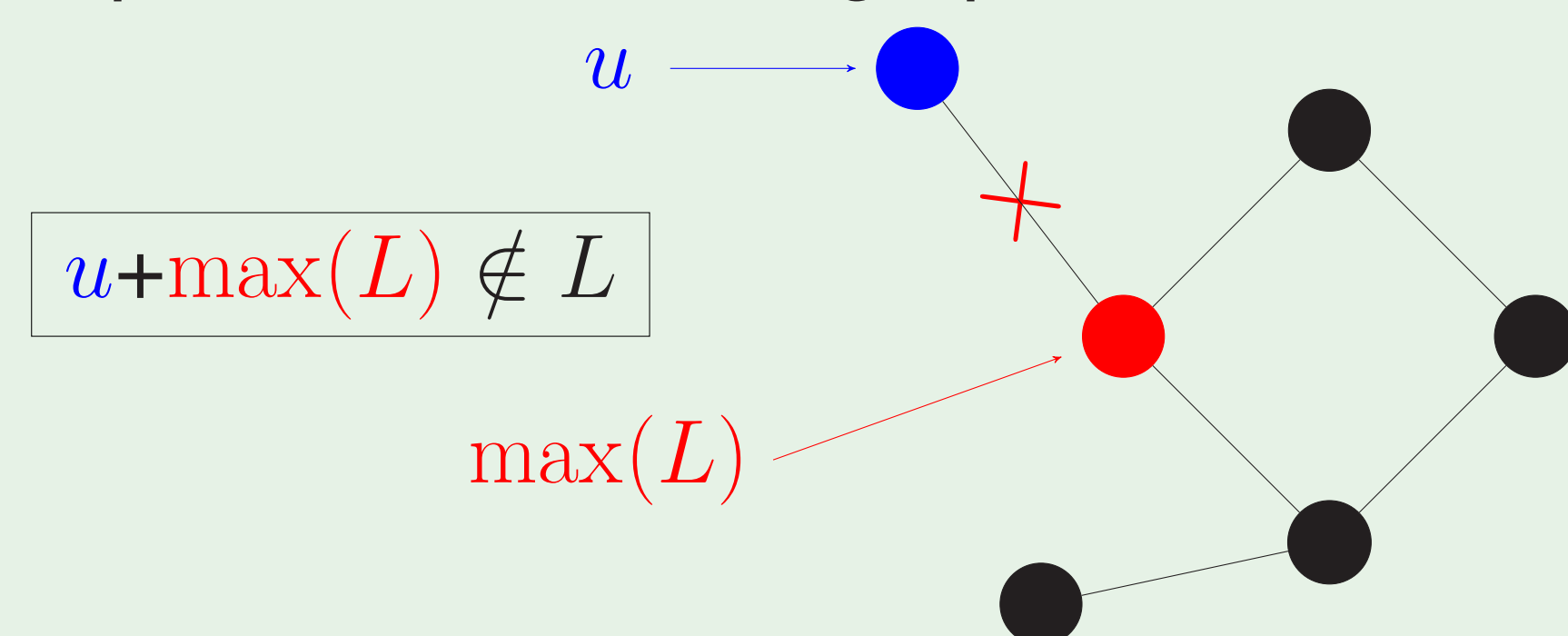
- MIT PRIMES-USA
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- Felix Gotti
- Tanya Khovanova
- Pavel Etingof
- AMS/PME/JMM

References

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- S. Singla, A. Tiwari, and A. Tripathi. Some results on the spum and the integral spum of graphs. *Discrete Math.* 344 (2021), P3.32.

Is every graph a sum graph?

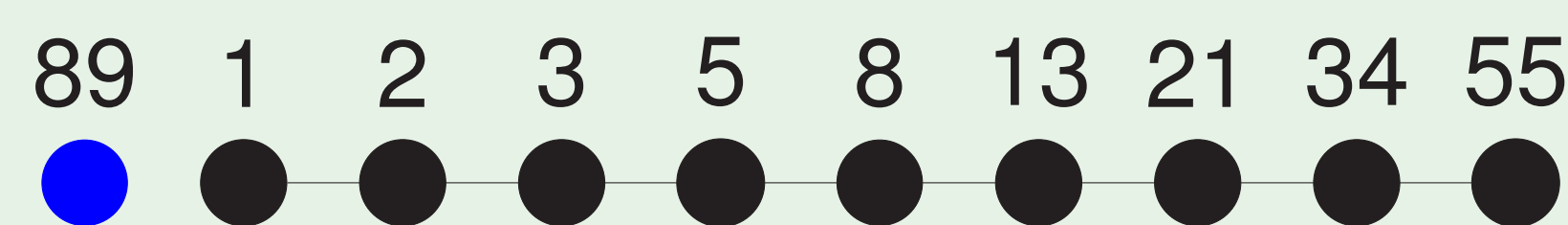
- No.** For example, all connected graphs are NOT sum graphs.



Sum Number $\sigma(G)$ (Harary '90)

- For any graph G , define $\sigma(G)$ to be the minimum number such that the union of G and $\sigma(G)$ isolated vertices is a sum graph.
- For paths P_n with n vertices, $\sigma(P_n) = 1$.

Example: $\sigma(P_9) = 1$



Spum $\text{spum}(G)$ (Goodell et al. '90)

- Let $\text{spum}(G)$ be the minimum $\text{range}(L)$ over all sum graph labelings L of the union of graph G and $\sigma(G)$ isolated vertices.
- $\text{spum}(C_n)$, $\text{spum}(K_n)$, and $\text{spum}(K_{1,n})$ are known but not $\text{spum}(P_n)$.

Theorem 1: Spum

Theorem (Li, '22)

- If $n \geq 8$, then

$$\text{spum}(P_n) \in \begin{cases} [2n - 2, 2n - 1] & \text{if } n \text{ is even} \\ [2n - 2, 2n + 1] & \text{if } n \text{ is odd} \end{cases}$$

Theorem (B.C.T., '23)

- If $n \geq 8$, then

$$\text{spum}(P_n) = \begin{cases} 2n - 1 & \text{if } n \text{ is even} \\ 2n + 1 & \text{if } n \text{ is odd} \end{cases}$$